DAY 08 – NETWORK CHALLENGE

Network Identity, DNS Investigation & Network Path Investigation

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Assignment	StudentName_Day08_Challenge.pdf

TASK 1 — NETWORK IDENTITY (8 Marks)

The following details are transcribed from the supplied Windows ipconfig /all screenshots. The full-detail screenshot shows the Wi-Fi adapter using the 10.219.5.0/24 network.

Required detail	Observed value
Active adapter	Wireless LAN adapter Wi-Fi (Realtek RTL8822CE 802.11ac PCIe Adapter)
IPv4 address	10.219.5.74
Subnet mask	255.255.255.0
Default gateway	10.219.5.52
DNS server	10.219.5.52
Physical (MAC) address	F8-89-D2-B0-CA-7D
DHCP enabled	Yes

Screenshot discrepancy to verify: Another supplied ipconfig screenshot shows the Wi-Fi IPv4 address as 10.72.32.74 and the default gateway as 10.72.32.141. This indicates that the computer was connected to a different network or had a different DHCP lease when that screenshot was taken. Use one fresh ipconfig /all capture for the final submission and make sure all Task 1 values come from that same capture.

TASK 4 — DNS INVESTIGATION (6 Marks)

Command to run in Windows Command Prompt:

```
nslookup google.com
```

What to record from the command output:

- DNS server that responded: _____
- IP address(es) returned for google.com: _____

Purpose of DNS: The Domain Name System (DNS) translates human-readable domain names, such as google.com, into IP addresses that computers use to locate and communicate with servers. The DNS server shown by nslookup is the resolver that answered the query; the returned address may vary by location, time, or DNS provider.

[Insert your nslookup google.com screenshot here]

Screenshot status: Not included in the files supplied for this report.

TASK 5 — NETWORK PATH INVESTIGATION (6 Marks)

Command to run in Windows Command Prompt:

```
tracert google.com
```

What to record from the command output:

- Number of visible hops: _____
- First hop IP address: _____
- Device commonly represented by the first hop: _____

What traceroute is used for: Traceroute (tracert in Windows) displays the sequence of network hops that packets pass through on the way to a destination. It helps identify the route taken and can assist in locating delays or connectivity problems. The first hop commonly represents the local default gateway/router, although network design can vary.

[Insert your tracert google.com screenshot here]

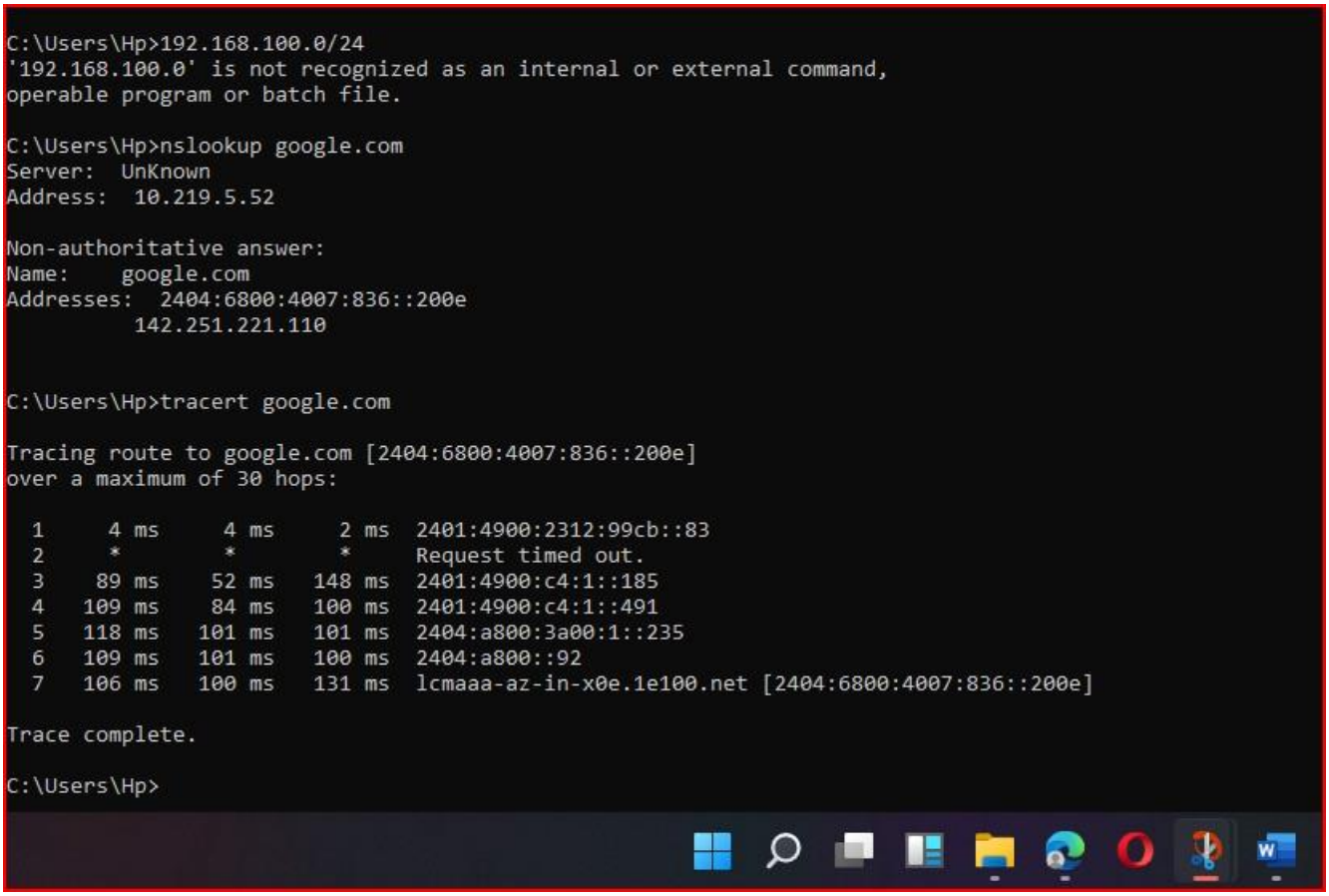
Screenshot status: Not included in the files supplied for this report.

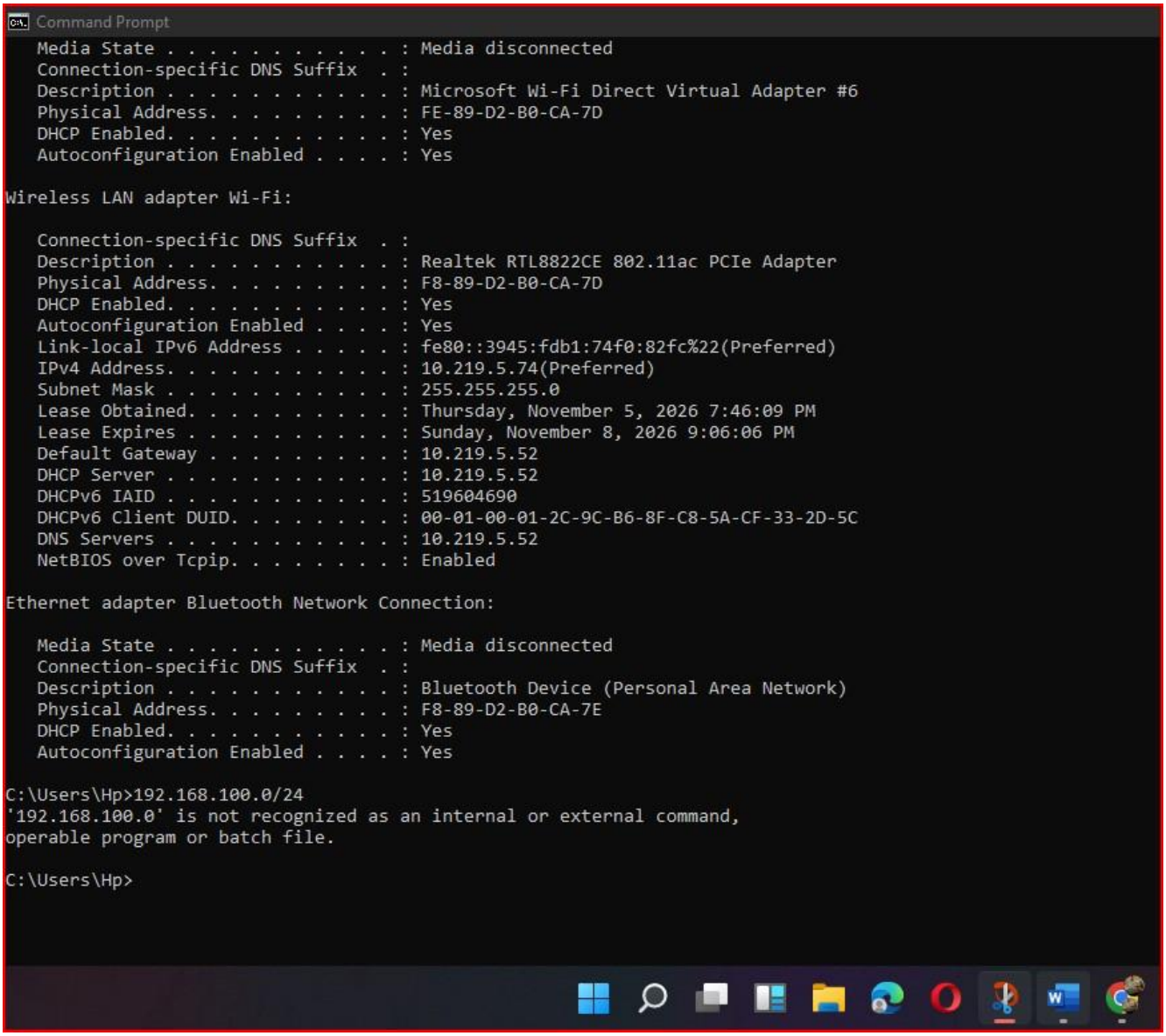
Commands Used

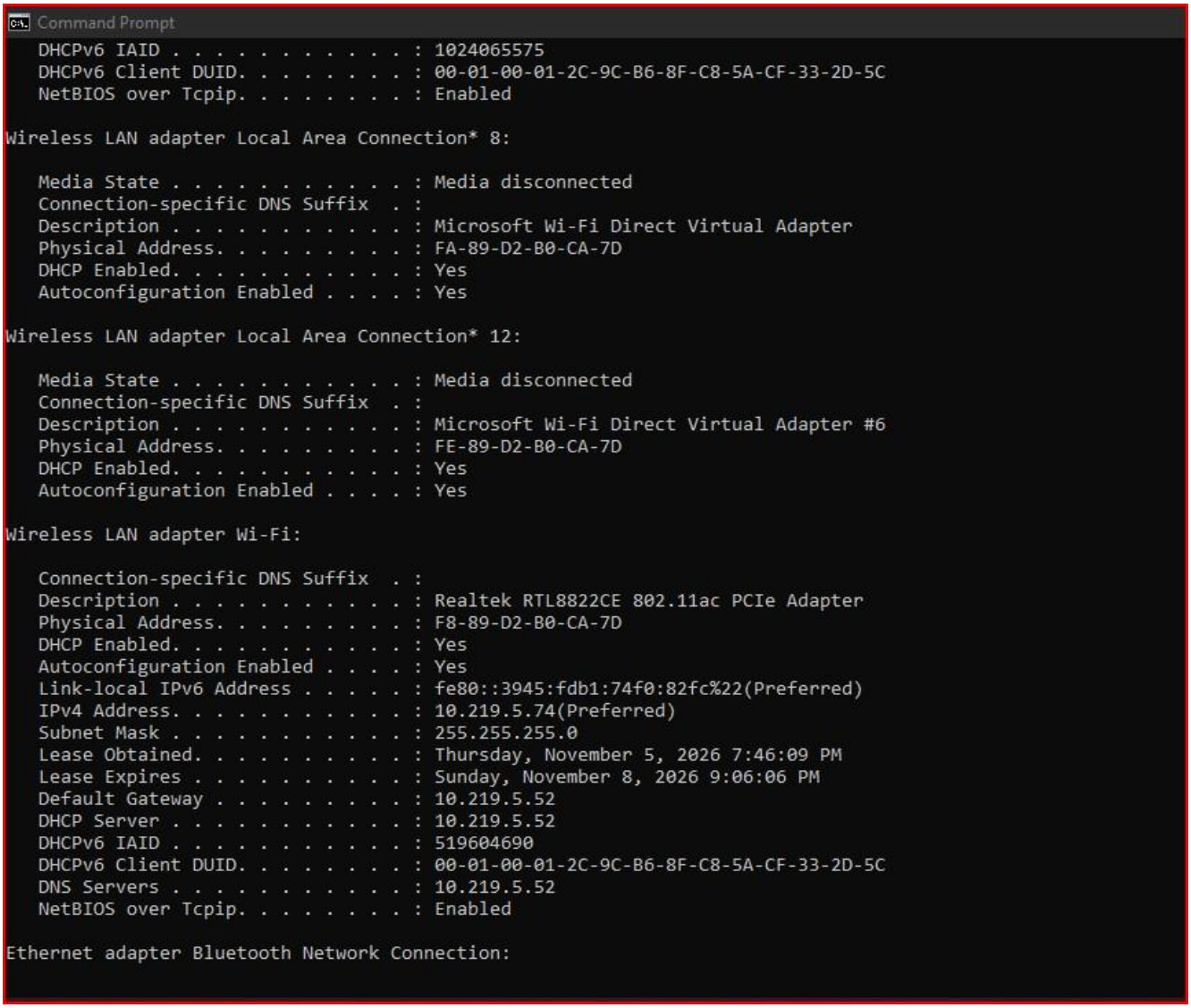
`ipconfig /all` — Displays detailed IP configuration, including adapter information, DHCP, gateway, DNS, and physical address.

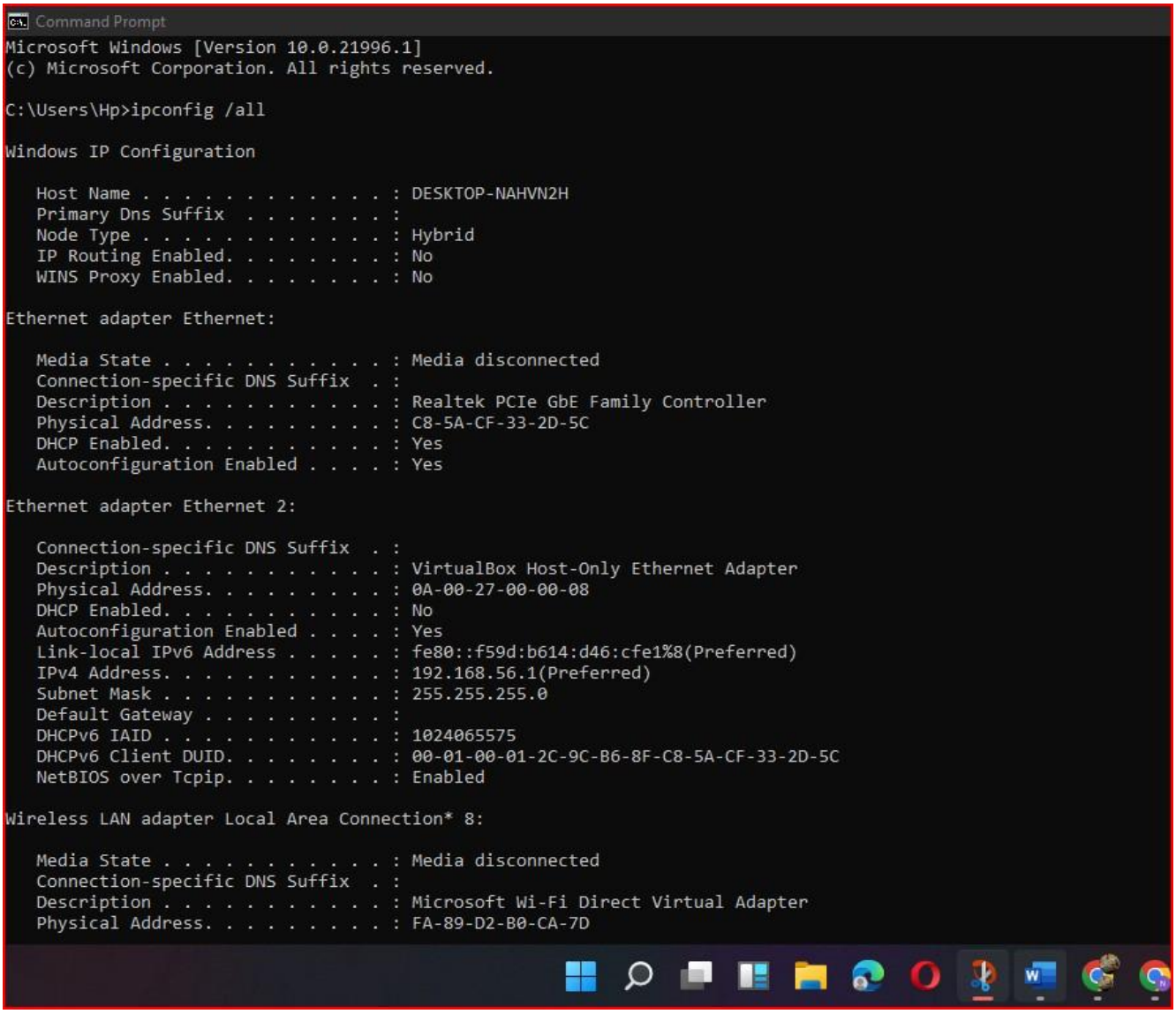
`nslookup google.com` — Queries DNS to find the IP address associated with the domain name.

`tracert google.com` — Displays the route and intermediate hops to the destination.


C:\Users\Hp>192.168.100.0/24
'192.168.100.0' is not recognized as an internal or external command,
operable program or batch file.
C:\Users\Hp>nslookup google.com
Server: UnKnown
Address: 10.219.5.52
Non-authoritative answer:
Name: google.com
Addresses: 2404:6800:4007:836::200e
142.251.221.110
C:\Users\Hp>tracert google.com
Tracing route to google.com [2404:6800:4007:836::200e]
over a maximum of 30 hops:
1 4 ms 4 ms 2 ms 2401:4900:2312:99cb::83
2 * * * Request timed out.
3 89 ms 52 ms 148 ms 2401:4900:c4:1::185
4 109 ms 84 ms 100 ms 2401:4900:c4:1::491
5 118 ms 101 ms 101 ms 2404:a800:3a00:1::235
6 109 ms 101 ms 100 ms 2404:a800::92
7 106 ms 100 ms 131 ms lcmaaa-az-in-x0e.1e100.net [2404:6800:4007:836::200e]
Trace complete.
C:\Users\Hp>


Command Prompt
Media State . . . . . : Media disconnected
Connection-specific DNS Suffix . :
Description . . . . . : Microsoft Wi-Fi Direct Virtual Adapter #6
Physical Address. . . . . : FE-89-D2-B0-CA-7D
DHCP Enabled. . . . . : Yes
Autoconfiguration Enabled . . . . : Yes
Wireless LAN adapter Wi-Fi:
Connection-specific DNS Suffix . :
Description . . . . . : Realtek RTL8822CE 802.11ac PCIe Adapter
Physical Address. . . . . : F8-89-D2-B0-CA-7D
DHCP Enabled. . . . . : Yes
Autoconfiguration Enabled . . . . : Yes
Link-local IPv6 Address . . . . . : fe80::3945:fdb1:74f0:82fc%22(Preferred)
IPv4 Address. . . . . : 10.219.5.74(Preferred)
Subnet Mask . . . . . : 255.255.255.0
Lease Obtained. . . . . : Thursday, November 5, 2026 7:46:09 PM
Lease Expires . . . . . : Sunday, November 8, 2026 9:06:06 PM
Default Gateway . . . . . : 10.219.5.52
DHCP Server . . . . . : 10.219.5.52
DHCPv6 IAID . . . . . : 519604690
DHCPv6 Client DUID. . . . . : 00-01-00-01-2C-9C-B6-8F-C8-5A-CF-33-2D-5C
DNS Servers . . . . . : 10.219.5.52
NetBIOS over Tcpip. . . . . : Enabled
Ethernet adapter Bluetooth Network Connection:
Media State . . . . . : Media disconnected
Connection-specific DNS Suffix . :
Description . . . . . : Bluetooth Device (Personal Area Network)
Physical Address. . . . . : F8-89-D2-B0-CA-7E
DHCP Enabled. . . . . : Yes
Autoconfiguration Enabled . . . . : Yes
C:\Users\Hp>192.168.100.0/24
'192.168.100.0' is not recognized as an internal or external command,
operable program or batch file.
C:\Users\Hp>


Command Prompt
DHCPv6 IAID . . . . . : 1024065575
DHCPv6 Client DUID. . . . . : 00-01-00-01-2C-9C-B6-8F-C8-5A-CF-33-2D-5C
NetBIOS over Tcpip. . . . . : Enabled
Wireless LAN adapter Local Area Connection* 8:
Media State . . . . . : Media disconnected
Connection-specific DNS Suffix . :
Description . . . . . : Microsoft Wi-Fi Direct Virtual Adapter
Physical Address. . . . . : FA-89-D2-B0-CA-7D
DHCP Enabled. . . . . : Yes
Autoconfiguration Enabled . . . . : Yes
Wireless LAN adapter Local Area Connection* 12:
Media State . . . . . : Media disconnected
Connection-specific DNS Suffix . :
Description . . . . . : Microsoft Wi-Fi Direct Virtual Adapter #6
Physical Address. . . . . : FE-89-D2-B0-CA-7D
DHCP Enabled. . . . . : Yes
Autoconfiguration Enabled . . . . : Yes
Wireless LAN adapter Wi-Fi:
Connection-specific DNS Suffix . :
Description . . . . . : Realtek RTL8822CE 802.11ac PCIe Adapter
Physical Address. . . . . : F8-89-D2-B0-CA-7D
DHCP Enabled. . . . . : Yes
Autoconfiguration Enabled . . . . : Yes
Link-local IPv6 Address . . . . . : fe80::3945:fdb1:74f0:82fc%22(Preferred)
IPv4 Address. . . . . : 10.219.5.74(Preferred)
Subnet Mask . . . . . : 255.255.255.0
Lease Obtained. . . . . : Thursday, November 5, 2026 7:46:09 PM
Lease Expires . . . . . : Sunday, November 8, 2026 9:06:06 PM
Default Gateway . . . . . : 10.219.5.52
DHCP Server . . . . . : 10.219.5.52
DHCPv6 IAID . . . . . : 519604690
DHCPv6 Client DUID. . . . . : 00-01-00-01-2C-9C-B6-8F-C8-5A-CF-33-2D-5C
DNS Servers . . . . . : 10.219.5.52
NetBIOS over Tcpip. . . . . : Enabled
Ethernet adapter Bluetooth Network Connection:



Conclusion

This challenge develops practical understanding of a computer's network identity, DNS name resolution, and packet routing across networks. These skills are useful in network troubleshooting and cybersecurity because they help identify hosts, understand name resolution, and investigate the path network traffic follows.